Principles of Economics

Final termTopic 2

Measuring the Cost of Living – Consumer Price Index (CPI)

THE CONSUMER PRICE INDEX

- The *consumer price index (CPI)* is a measure of the overall cost of the goods and services bought by a typical consumer.

- CPI is used to monitor changes in the cost of living over time.

THE CONSUMER PRICE INDEX

- When the CPI rises, the typical family has to spend more money to maintain the same standard of living.

How the Consumer Price Index Is Calculated

- ***Fix the Basket:*** Determine what prices are most important to the typical consumer.
  - The Economic Agency identifies a basket of goods and services the typical consumer buys.

Step 1: Survey Consumers to Determine a Fixed Basket of Goods

4 hot dogs, 2 hamburgers

How the Consumer Price Index Is Calculated

- ***Find the Prices:*** Find the prices of each of the goods and services in the basket for each point in time.

Step 2: Find the Price of Each Good in Each Year

Year	Price of Hot Dogs	Price of Hamburgers
2001	$1	$2
2002	2	3
2003	3	4

How the Consumer Price Index Is Calculated

- ***Compute the Basket's Cost:*** Use the data on prices to calculate the cost of the basket of goods and services at different times.

Step 3: Compute the Cost of the Basket of Goods in Each Year

2001	($1 per hot dog × 4 hot dogs) + ($2 per hamburger × 2 hamburgers) = $8
2002	($2 per hot dog × 4 hot dogs) + ($3 per hamburger × 2 hamburgers) = $14
2003	($3 per hot dog × 4 hot dogs) + ($4 per hamburger × 2 hamburgers) = $20

- CPI = Cost of Basket in the current year / cost of basket in the base year *100
- Assume : 2001 is the base year
- Cost of basket 2001 : $8
- Cost of Basket 2002: $14
- Cost of Basket 2003:$20

CPI 2001 = (8/8) *100 = 100

CPI 2002 = (14/8) *100 = 175

CPI 2003 = (20/8) *100 = 250

How the Consumer Price Index Is Calculated

- ***Choose a Base Year and Compute the Index:***
  - Designate one year as the base year, making it the benchmark against which other years are compared.
  - Compute the index by dividing the price of the basket in one year by the price in the base year and multiplying by 100.

Step 4: Choose One Year as a Base Year (2001) and Compute the Consumer Price Index in Each Year

2001	($8/$8) × 100 = 100
2002	($14/$8) × 100 = 175
2003	($20/$8) × 100 = 250

How the Consumer Price Index Is Calculated

- ***Compute the inflation rate:*** The inflation rate is the percentage change in the price index from the preceding period.

How the Consumer Price Index Is Calculated

- The Inflation Rate
  - The *inflation rate* is calculated as follows:

$$\text{Inflation Rate in Year 2} = \frac{\text{CPI in Year 2} - \text{CPI in Year 1}}{\text{CPI in Year 1}} \times 100$$

Step 5: Use the Consumer Price Index to Compute the Inflation Rate from Previous Year

2002	(175 − 100)/100 × 100 = 75%
2003	(250 − 175)/175 × 100 = 43%

CPI Vs GDP deflator

GDP deflator & CPI give some what different information about what's happening to the overall level of prices in the economy.

Three Key Differences:

1. GDP deflator measures the prices of all goods and services produced whereas CPI measures the prices of only the goods and services bought by consumers.

2. GDP deflator includes only those goods produced domestically. Imported goods are not a part of GDP and do not show up in the GDP deflator.

3. CPI include imported goods.CPI is computed using a fixed basket of goods whereas GDP deflator allows the basket of goods to change over time as the composition of goods and services changed

Problems in Measuring the Cost of Living-CPI Biases

- The CPI is an accurate measure of the selected goods that make up the typical bundle, but it is not a perfect measure of the cost of living.

Problems in Measuring the Cost of Living

- Substitution bias
- Introduction of new goods

Problems in Measuring the Cost of Living-CPI Biases

- Substitution Bias
  - The basket does not change to reflect consumer reaction to changes in relative prices.
    - Consumers substitute toward goods that have become relatively less expensive.
    - The index overstates the increase in cost of living by not considering consumer substitution.

Problems in Measuring the Cost of Living

- Introduction of New Goods
  - The basket does not reflect the change in purchasing power brought on by the introduction of new products.
    - New products result in greater variety, which in turn makes each dollar more valuable.
    - Consumers need fewer dollars to maintain any given standard of living.

Problems in Measuring the Cost of Living-CPI Biases

- The substitution bias, introduction of new goods, cause the CPI to overstate the true cost of living.
  - The issue is important because many government programs use the CPI to adjust for changes in the overall level of prices.

Nominal & Real Interest rate

The *nominal interest* rate is the interest rate usually reported and not corrected for inflation. It is the interest rate that a bank pays.

The *real interest rate* is the nominal interest rate that is corrected for the effects of inflation.

- Real Interest rate = Nominal interest rate – Inflation rate

INFLATION

- **Inflation** refers to a situation in which the economy's overall price level is rising.
- The **inflation rate** is the percentage change in the price level from the previous period.
  - **Not all prices rise at the same rate during inflation.**
  - **Not everyone suffers equally from inflation.**
  - **Although inflation makes some people worse off, it makes some people better off**
- **Hyperinflation is an extraordinarily high rate of inflation such as Germany experienced in the 1920s.**
- **Hyperinflation is inflation that exceeds 50% per month**

Types of Inflation

- Demand-Pull Inflation
  - Demand-pull inflation results from excessive pressure on the demand side of the economy.
  - “Too much money chases too few goods” enabling producers to raise prices.
- Cost-Push Inflation
  - The pressure on price could also originate on the supply side.
  - Higher production costs put upward pressure on product prices

Effects of Inflations [on..]

	Consumers	Producers	Economy
Zero inflation	Not affected at all	No incentive to produce more, Same production	Stagnant for almost same level of production
Mild Inflation	Affected but not much, Demand may be same	Have incentive to produce more, Higher production	Economy expands as production increase
High Inflation	Affected much, Demand may be lower in many products	Producers of inelastic products affected, Production lower for lower demand	Economy squeezes as production falls

Summary

- The consumer price index shows the cost of a basket of goods and services relative to the cost of the same basket in the base year.
- The index is used to measure the overall level of prices in the economy.
- The percentage change in the CPI measures the inflation rate.

Summary

- The consumer price index is an imperfect measure of the cost of living for the following three reasons: substitution bias, and the introduction of new goods.